

Advanced Microeconomics

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Winter term 2025/26

Problem Set 6: Application of income and substitution effects and welfare measures

1 Application

1) A consumer's utility function is $u = x_1^{1/2} x_2^{1/2}$. She has an income I and prices are p_1 and p_2 . Explore the relationship between utility maximization and expenditure minimization while filling out the values for the given utility function in the table.

Utility Maximizing	Expenditure Minimizing
Marshallian demands: $x_1 = I/2p_1$	Hicksian demands: $h_1 = u(p_2/p_1)^{1/2}$

Plug in the respective demand functions to get the

indirect utility function: $v = I/(4p_1p_2)^{1/2}$	expenditure function: $e = u(4p_1p_2)^{1/2}$
Substitute the expenditure function (e) into the Marshallian demand function to derive the Hicksian demand function: $x_1 = (u(4p_1p_2)^{1/2})/2p_1 = u(p_2/p_1)^{1/2}$	Substitute the indirect utility function (v) into the Hicksian demand function to derive the Marshallian demand function: $h_1 = ((p_2/p_1)^{1/2})(I/(4p_1p_2)^{1/2}) = I/2p_1$
Invert v and replace I by u to get the expenditure function: $v^{-1} = u(4p_1p_2)^{1/2}$	Invert e and replace u by v to get the indirect utility function: $e^{-1} = I(4p_1p_2)^{-1/2}$
Check Roy's identity: $x_1 = -\frac{\frac{\partial v}{\partial p_1}}{\frac{\partial v}{\partial I}} = \frac{2I(p_1p_2)^{1/2}}{4(p_1^3p_2)^{1/2}} = \frac{I}{2p_1}$	Check Shephard's lemma: $h_1 = \frac{\partial e}{\partial p_1} = \frac{u4p_2}{2(4p_1p_2)^{1/2}} = u(p_2/p_1)^{1/2}$

Establish the Slutsky equation: $\frac{\partial x_1}{\partial p_2} = \frac{u}{2(p_1p_2)^{1/2}} - \frac{I}{2p_2} \cdot \frac{1}{2p_1}$

substitute $u = v(p, I)$ into the substitution effect: $\frac{\partial x_1}{\partial p_2} = \frac{I}{4p_1p_2} - \frac{I}{4p_1p_2} = 0$

2) Suppose there are two goods, prices are given, and there is a consumer with a certain positive income.

- a) Show in a diagram how it is possible to split the effect of a price decrease on good 1 into the income and substitutions effects. Assume that the good is a normal good.

Solution: We have a consumer who maximizes her utility at a certain point on the budget line (budget restriction). In that point the budget line must be a tangent to an indifference curve. We therefore have an indifference curve such as I_1 and a budget line such as BL_1 in Figure 1. The point of utility maximization is point A.

If the price of good 1 is cut by half the consumer can maximally buy twice the amount she could before. The budget line therefore rotates outwards to BL_2 and intersects the X-axis at twice the distance.

Suppose that the consumer buys more of good 1 if the price falls. The new point of utility maximization must therefore be on the right of point A. For instance, it could be at point B. The total effect of the price fall is that the consumer shifts from consuming the quantity q_{11} of good 1 to the quantity q_{12} .

We can split the total effect into the income and substitution effect. The substitution effect is the change in demand on good 1 that only depends on the change in relative prices and not the increase in the level of utility. To find the size of that effect we construct an imaginary budget line BL_* . Two things are important with BL_* . First, the relative prices must be the same as after the price decrease, implying that BL_* must have the same slope as BL_2 . Second, the level of utility at the point of utility maximization on BL_* must be the same as the consumer originally had (since the substitution effect does not measure the increase in utility). BL_* must consequently be a tangent to the same indifference curve to which the original budget line, BL_1 , is a tangent I_1 .

However, since BL_* and BL_1 have different slopes, the point of tangency will not be the same for both of them. For BL_1 it is point A. For BL_* it is point C. From the graph, we see directly that the substitution effect must be positive for price decreases and negative for price increases. If BL_* has a slope that is smaller than that of BL_1 , the point of tangency must be to the right of the previous point. (Note however that the substitution effect can be zero)

The substitution effect is the increase in consumption of good 1 that corresponds to the distance between point A and C along the X-axis. The consumer has the same level of utility but consumes more of good 1 since it is cheaper than before in relative terms. Instead, she consumes less of good 2. Note that this is not something you can observe. BL_* and point C are theoretical constructs.

The income effect is the part of the total effect that remains after the substitution effect has been subtracted. It corresponds to the part of the change in demand of good 1 that depends on the fact that the consumer increases her utility. In this case she moves from the indifference curve I_1 to I_2 . If BL_* would have been a real budget line, an increase in income that had shifted BL_* to BL_2 would have given the same increase in the demand of good 1.

- b) If the good had been an inferior good, what would have been different in the graph.

Solution: An inferior good is a good one buys less of when income increases, since one shifts over to other goods of higher quality. Regarding the substitution and income effects, the inferior good must have a negative income effect. In Figure 1 that would mean point B would have been somewhere left of C instead. To construct such a graph, draw the indifference curve such that BL_2 is a tangent to it to the left of C, for instance as I_3 in the graph.

- c) If the good had been an Giffen good, what would have been different.

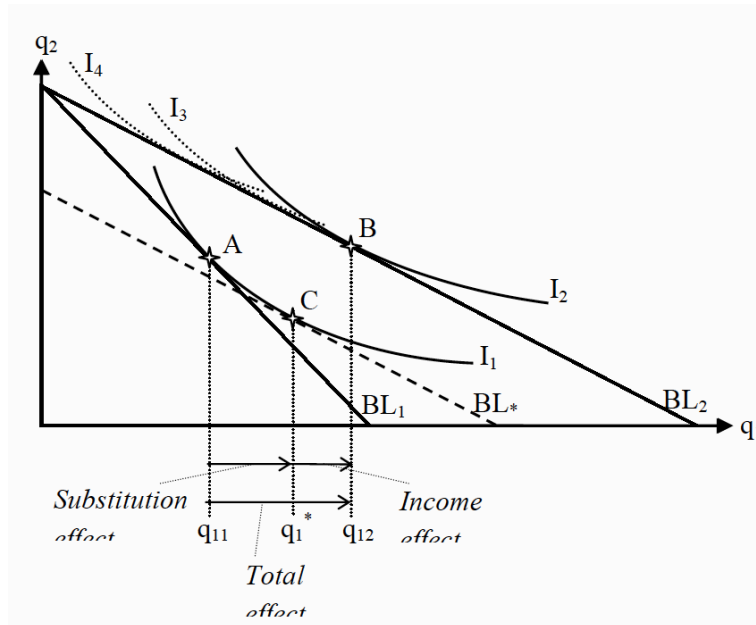


Figure 1: Solution for a)

Solution: A Giffen good is a special type of inferior good. The income effect is so large that it dominates over the substitution effect. To construct such a graph, draw the indifference curve such that BL_2 is a tangent to it to the left of point A.

3) A consumer has preferences over the single good x and all other goods m represented by the utility function, $u(x, m) = \ln(x) + m$. Let the price of x be p , the price of m be unity, and let income be I .

a) Derive the Marshallian demands for x and m .

Solution: The equality of marginal rate of substitution and price ratio gives $1/x = p$. Thus, the Marshallian demand for x is $x = 1/p$. The uncompensated demand for m separates into two cases depending on the amount of income available:

$$m = \begin{cases} 0 & \text{when } I \leq 1 \\ I - 1 & \text{when } I > 1 \end{cases}$$

b) Derive the indirect utility function, $v(p, I)$.

Solution: Again, depending on the amount of income available there will be two indirect utility functions:

$$v(p, I) = \begin{cases} \ln(\frac{1}{p}) & \text{when } m \leq 1 \\ I - 1 - \ln(p) & \text{when } m > 1 \end{cases}$$

c) Use the Slutsky equation to decompose the effect of an own-price change on the demand for x into an income and substitution effect. Interpret your result briefly.

Solution: A well-known property of any demand function derived from a quasi-linear utility function is the absence of the income effect. Which can be easily seen in the application of the Slutsky equation:

$$\frac{\partial x^h}{\partial p} = \frac{\partial x}{\partial p} + x \frac{\partial x}{\partial I}$$

$$\frac{\partial x}{\partial p} = -\frac{1}{p^2} + 0 \frac{1}{p} = \frac{\partial x^h}{\partial p}$$

Therefore, the effect of an own-price change on the demand for x equals the substitution effect.

- d) Suppose that the price of x rises from p^0 to $p^1 > p^0$. Show that the consumer surplus area between p^0 and p^1 gives an exact measure of the effect of the price change on consumer welfare.

Solution: The consumer surplus area can be calculated by integrating over the inverse demand function of x:

$$CS = \int_{p^0}^{p^1} \frac{1}{x} dx = \ln(p^1 - p^0)$$

Calculating the change in utility induced by a price change gives:

$$\Delta v = v^1(p^1, I^1) - v^0(p^1, I^0) = I - 1 - \ln(p^1) - (I - 1 - \ln(p^0)) = \ln(p^1 - p^0)$$

As the two expressions are equal, the consumer surplus area gives an exact measure of the effect of the price change on consumer welfare in the case of quasi-linear preferences.

- e) Carefully illustrate your findings with a set of two diagrams: one giving the indifference curves and budget constraints on top, and the other giving the Marshallian and Hicksian demands below. Be certain that your diagrams reflect all qualitative information on preferences and demands that you've uncovered. Be sure to consider the two prices p^0 and p^1 , and identify the Hicksian and Marshallian demands.

Solution: See Figure 2. Please note, that Hicksian and Marshallian demands are identical here.

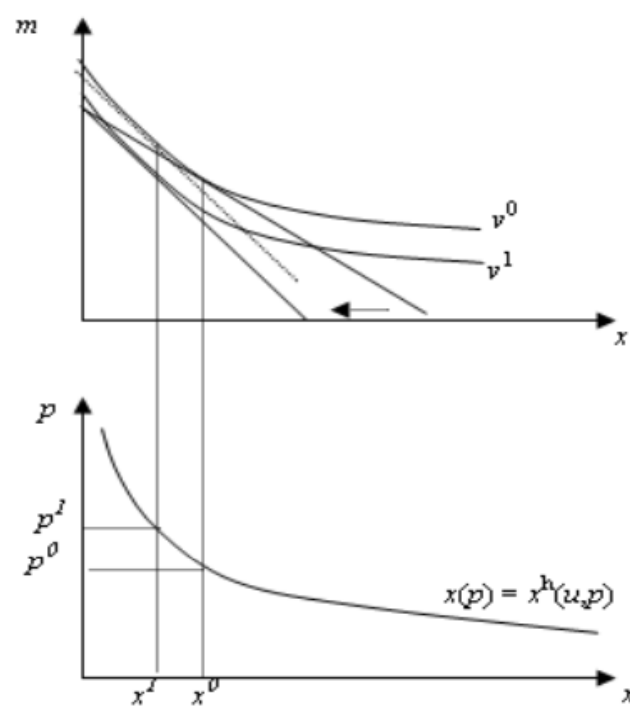


Figure 2: Solution for e)